

The empirical recurrence for OEIS A250951, proved by transfer matrix

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Abstract

OEIS A250951 counts $(n+1) \times 4$ arrays over $\{0, \dots, 1\}$ subject to a condition imposed on every 2×2 subblock, and carries an empirical recurrence of order 3 contributed by R. H. Hardin. It is true, and it is not an empirical matter at all. The condition is local to each subblock, so the rows of an admissible array are the vertices of a finite digraph on $S = 16$ vertices whose edges are the admissible consecutive pairs, and the count is a number of walks: $a(n) = \mathbf{1}^\top M^n \mathbf{1}$. Writing q for the characteristic polynomial of the conjectured recurrence, the recurrence holds for all n exactly when $\mathbf{1}^\top M^j q(M) \mathbf{1} = 0$ for every $j \geq 0$, and the Cayley–Hamilton theorem bounds that to $j < S$. It is therefore a finite computation in exact integer arithmetic, and it comes out zero.

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1 The sequence and the conjecture

OEIS A250951 is “Number of $(n+1) \times (3+1)$ 0..1 arrays with no 2×2 subblock having its maximum diagonal element less than its minimum antidiagonal element.”. It has offset 1 and begins

209, 2744, 36016, 472712, 6204344, 81431944, 1068793208, 14027896712, ...

The entry carries the following comment:

Empirical: $a(n) = 15*a(n-1) - 24*a(n-2) - 8*a(n-3)$.

As of the “Last modified” line on the live entry (Nov 24 2018 08:11:25, revision 10) the statement is still recorded as empirical, and nothing on the entry records it as settled either way.

Written out, the claim is

$$a(n) = 15a(n-1) - 24a(n-2) - 8a(n-3) \tag{1}$$

for all $n > 2$.

2 The condition is local, so the rows form a digraph

Write an array of the shape in question as its list of rows $r^{(1)}, r^{(2)}, \dots$, each an element of $V = \{0, 1, \dots, 1\}^4$, so $|V| = S = 16$. A 2×2 subblock of the array occupies two consecutive rows and two consecutive columns; if the two rows are r and s (in that order) and the two columns are numbered $j, j+1$, the subblock is

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} r_j & r_{j+1} \\ s_j & s_{j+1} \end{pmatrix}.$$

The entry's requirement on that subblock is

$$\text{not } (\max(\alpha, \delta) < \min(\beta, \gamma)), \quad (2)$$

a condition on the four entries alone.

For $r, s \in V$ declare $r \rightarrow s$ an edge of a digraph G when, for every $1 \leq j \leq 4 - 1$,

- the four entries above satisfy (2);

Let $M \in \{0, 1\}^{S \times S}$ be the adjacency matrix of G and $\mathbf{1}$ the all-ones vector.

Lemma 1. *The arrays counted by the entry with L rows are exactly the walks $r^{(1)} \rightarrow r^{(2)} \rightarrow \dots \rightarrow r^{(L)}$ of length $L - 1$ in G . Consequently*

$$a(n) = \mathbf{1}^\top M^n \mathbf{1}.$$

Proof. An array with L rows and 4 columns is precisely a sequence $r^{(1)}, \dots, r^{(L)}$ of elements of V ; conversely any such sequence is an array. Every 2×2 subblock lies in two consecutive rows $r^{(t)}, r^{(t+1)}$ and two consecutive columns, so the array satisfies the entry's condition on all of its subblocks if and only if each consecutive pair $(r^{(t)}, r^{(t+1)})$ satisfies it for every column pair — which is exactly the edge relation of G . The number of walks of length $L - 1$, summed over all starting and ending vertices, is $\mathbf{1}^\top M^{L-1} \mathbf{1}$, since $(M^k)_{r,s}$ counts the walks of length k from r to s . The entry's index n corresponds to $L = n + 1$ rows. $\square$

3 The criterion

Let $q(t) = t^3 - \sum_i c_i t^{3-i}$ be the characteristic polynomial of (1), with the c_i read off that equation.

Lemma 2. *For every n with $n + 1 - 1 \geq 3$,*

$$a(n) - \sum_i c_i a(n - i) = \mathbf{1}^\top M^{n+1-1-3} q(M) \mathbf{1}.$$

Hence (1) holds from that point on if and only if $\mathbf{1}^\top M^j q(M) \mathbf{1} = 0$ for every $j \geq 0$; and it is enough to check $j = 0, 1, \dots, S - 1$.

Proof. By Lemma 1, $a(n - i) = \mathbf{1}^\top M^{n+1-1-i} \mathbf{1}$, so

$$a(n) - \sum_i c_i a(n - i) = \mathbf{1}^\top \left(M^m - \sum_i c_i M^{m-i} \right) \mathbf{1} = \mathbf{1}^\top M^{m-3} q(M) \mathbf{1}$$

with $m = n + 1 - 1$, which is the stated identity; the equivalence follows by letting $m - 3 = j$ run over ≥ 0 . For the last clause put $w = q(M) \mathbf{1}$. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so $\mathbf{1}^\top M^j w$ for $j \geq S$ is the corresponding combination of the earlier values; if those all vanish, so do all the rest. $\square$

4 The computation

Theorem 3. *The recurrence (1) holds for every $n > 2$, which is the statement of Section 1.*

Proof. The digraph was constructed from (2) by a depth-first scan across the 4 columns: the edge condition is a conjunction of one constraint per consecutive column pair, so the successors of a row r are enumerated column by column without testing all S^2 pairs. The vector $w = q(M) \mathbf{1} \in \mathbb{Z}^S$ was then formed by 3 matrix–vector products in exact integer arithmetic, and $\mathbf{1}^\top M^j w$ was evaluated for $j = 0, 1, \dots$, again exactly. Every one of those integers vanishes from the index corresponding to $n = 2 + 1$ onwards, and the run continued until w itself was the zero vector, which settles every larger j at once. By Lemma 2 the recurrence holds for every $n > 2$. $\square$

Remark 4. The argument gives more than the recurrence: it shows the sequence is C -finite with characteristic polynomial dividing that of M , so it has a rational generating function whose denominator divides $\det(I - xM)$. The conjectured recurrence is one consequence; the minimal one is a divisor of q .

5 Verification

Three checks, each independent of the algebra above.

First, the digraph was built directly from the entry's stated condition and the walk counts $\mathbf{1}^\top M^{L-1} \mathbf{1}$ compared against the entry's DATA at every one of the 15 published indices; the values agree exactly. That is what ties the digraph to the sequence: a model that reproduced only some of the published terms would be the wrong model, and would have been rejected. In particular it is what rules out a misreading of the entry's wording, since a different reading of the condition gives a different digraph and different counts.

Second, the recurrence (1) was evaluated on those published terms in exact integer arithmetic, with no matrices involved, and vanishes wherever it applies.

Third, the annihilation test of Lemma 2 was run in exact integer arithmetic until the working vector was identically zero, not to a sampled prefix, and returned zero throughout.

References

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- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
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- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.